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Concealed Carry

Contents

1 Big picture

- **Question.** Why did the traditional currency carry trade weaken after 2008 while a yield-curve slope carry strategy became much more profitable?
- **Traditional carry.** Borrow in low short-rate currencies and invest in high short-rate currencies. Equivalently, sort countries by the level of the short-term interest rate.
- **Slope carry.** Take long positions in long-term bonds of countries with steeper yield curves and short positions in countries with flatter yield curves.
- **Main empirical fact.** Traditional carry is positive over the full sample but weaker after 2008. Slope carry is close to zero over the full sample, slightly negative before 2008, and strongly positive after 2008.
- **Macro fact.** Expected global growth and expected global inflation both decline after the global financial crisis.
- **Country heterogeneity.** Countries differ in their exposure to news about global expected growth and global expected inflation.
- **Mechanism.** Traditional carry is mainly about heterogeneous exposure to global growth news. Slope carry is mainly about heterogeneous exposure to global inflation news.
- **Key intuition for slope carry.** When expected global inflation is below average, countries with high exposure to global inflation risk tend to have steeper yield curves and higher inflation risk premia. The slope carry then goes long those countries and earns a positive excess return.
- **Model contribution.** The paper embeds heterogeneous global growth and inflation exposures in a recursive-preference international asset-pricing model with complete markets.
- **Takeaway.** The apparent absence of a slope-carry premium in long samples is concealed by a negative pre-2008 premium and a strong positive post-2008 premium. The sign of the slope carry depends on the global inflation regime.

2 Data and measurement

2.1 Countries and yield-curve portfolios

- The empirical analysis focuses on G10 countries with heavily traded currencies and sovereign bond markets.
- For traditional carry, countries are sorted monthly into three GDP-weighted portfolios using the 3-month interest rate.
- Portfolio 1 contains low short-rate currencies; portfolio 3 contains high short-rate currencies.
- For slope carry, countries are sorted monthly into three GDP-weighted portfolios using the yield-curve slope, defined as the 10-year yield minus the 3-month yield.
- Portfolio 1 contains flatter yield curves; portfolio 3 contains steeper yield curves.

Interpretation: Traditional carry is a cross-section of short-rate levels. Slope carry is a cross-section of yield-curve slopes. The paper emphasizes that these two rankings are not equivalent.

2.2 Portfolio returns

The portfolio return for a bond strategy is a GDP-weighted return:

$$\log RF^X_{p,t} = \sum_{i \in p} w^p_{i,t} \log RF^X_{i,t}, \qquad p \in \{P1, P2, P3\}, \tag{1}$$

where the weights are

$$w_{i,t}^{p} = \frac{GDP_{i,t}}{\sum_{i \in p} GDP_{i,t}}. \tag{2}$$

The excess return of the high-minus-low strategy is

$$Carry_t = RF_{P3,t}^{X} - RF_{P1,t}^{X}. \tag{3}$$

Interpretation: The high-minus-low return isolates the payoff from the sorting variable: short-rate level for traditional carry, yield-curve slope for slope carry.

2.3 Exchange-rate component

The paper decomposes currency/bond returns into local bond returns and exchange-rate depreciation. For the extreme portfolios, the exchange-rate component is summarized as

$$E(\Delta FX)_{P3} - E(\Delta FX)_{P1}. \tag{4}$$

Interpretation: This decomposition clarifies whether carry returns come from bond holding-period returns or currency movements. For slope carry, exchange-rate depreciation contributes positively after 2008.

2.4 Global expectations

The paper constructs global expected growth and inflation as cross-country GDP-weighted averages:

$$E_t[\Delta y_{t+1}^{G10}] = \sum_i \omega_{i,t} E_t[\Delta y_{i,t+1}], \qquad E_t[\pi_{t+1}^{G10}] = \sum_i \omega_{i,t} E_t[\pi_{i,t+1}]. \tag{5}$$

- Expected GDP growth and expected inflation come from OECD forecasts.
- The paper splits the sample into pre- and post-2008 subperiods.
- Both expected growth and expected inflation fall after 2008.

Interpretation: The macro regime shift is central: after 2008, the world enters a lower expected inflation and lower expected growth environment.

2.5 Country exposures to global expectations

For each country, the paper estimates exposures to global expected growth and inflation:

$$E_t[\Delta y_{i,t+1}] = \mu_{i,y} + \beta_{i,y} E_t[\Delta y_{t+1}^{G10}] + \varepsilon_{i,t}, \tag{6}$$

$$E_t[\pi_{i,t+1}] = \mu_{i,\pi} + \beta_{i,\pi} E_t[\pi_{t+1}^{G10}] + \varepsilon_{i,t}. \tag{7}$$

Interpretation: These regressions quantify how sensitive each country's expected growth and inflation are to global expectations. Heterogeneous exposures are the key state variable for carry premia in the model.

3 Model and equilibrium conditions

3.1 Preferences

Each country is populated by a representative investor with recursive preferences:

$$U_{i,t} = (1-\delta)\log C_{i,t} + \delta \log E_t \exp\left\{\frac{U_{i,t+1}}{\theta}\right\}^{\theta}, \tag{8}$$

where risk aversion is γ, subjective discount factor is δ, and $\theta = 1/(1-\gamma)$ under unit intertemporal elasticity of substitution.

Interpretation: Recursive preferences generate risk premia for shocks to expected future growth and inflation.

3.2 Global long-run growth and inflation

The global state vector follows

$$\begin{bmatrix} x_{\pi,t} \\ x_{c,t} \end{bmatrix} = \begin{bmatrix} \rho_\pi & 0 \\ \rho_{c\pi} & \rho_c \end{bmatrix} \begin{bmatrix} x_{\pi,t-1} \\ x_{c,t-1} \end{bmatrix} + \begin{bmatrix} \sigma_{x\pi} & 0 \\ 0 & \sigma_{xc} \end{bmatrix} \begin{bmatrix} \varepsilon_{\pi,t} \\ \varepsilon_{c,t} \end{bmatrix}. \tag{9}$$

- $x_{c,t}$ is global expected consumption/output growth.
- $x_{\pi,t}$ is global expected inflation.
- $\rho_{c\pi}$ allows global inflation to affect expected global growth.

Interpretation: The model links global inflation news and global growth news. This allows slope carry and traditional carry to respond differently to macro regimes.

3.3 Country-level growth and inflation

Country-level consumption growth and inflation are

$$\Delta c_{i,t+1} = \mu_i^c + \beta_i^c x_{c,t} + \sigma_i^c \eta_{i,t+1}^c, \tag{10}$$

$$\pi_{i,t+1} = \mu_i^\pi + \beta_i^\pi x_{\pi,t} + \sigma_i^\pi \eta_{i,t+1}^\pi. \tag{11}$$

Interpretation: Countries differ in β_i^c and β_i^π. The traditional carry is driven mainly by β_i^c; the slope carry is driven mainly by β_i^π.

3.4 Slope carry risk premium

The conditional risk premium of the slope carry can be written as

$$E[Carry_t^S \mid x_{\pi,t}] = \log E_t[RF_{S,t+1}^{X,\infty}] - \log E_t[RF_{F,t+1}^{X,\infty}], \tag{12}$$

where S is the steeper-slope portfolio and F is the flatter-slope portfolio.

The sign of the spread is linked to expected inflation:

$$\mathrm{sign}(slope_{i,t}^{\infty} - slope_{j,t}^{\infty}) = -\mathrm{sign}(x_{\pi,t}). \tag{13}$$

Interpretation: When global expected inflation is below average, high-inflation-exposure countries have steeper yield curves and higher inflation-risk premia. The slope carry then earns a positive premium.

3.5 Entropy and portfolio composition

The log return to a currency-bond strategy can be decomposed using the entropy terms of the permanent components of stochastic discount factors:

$$E_t[\log RF_{i,t+1}^{X,\infty}] = TP_b + L_b - L_i. \tag{14}$$

For a portfolio,

$$E_t[\log RF_{p,t+1}^{X,\infty}] = \sum_{i \in P} \omega_{i,t}^p L_i + L_b + TP_b, \tag{15}$$

where portfolio weights change as countries move across the slope portfolios.

Interpretation: Country-level entropy is constant in the model, but portfolio-level entropy is time-varying because the portfolio composition changes across regimes.

4 Identification and empirical design

4.1 Main empirical design

- The paper compares traditional carry and slope carry over full, pre-2008, and post-2008 samples.
- It documents that the slope carry premium is concealed in the full sample by opposite signs across subperiods.
- It estimates country-level exposures to global expected growth and inflation.
- It shows that these exposures map into the recurring portfolio composition of traditional and slope carry strategies.

Interpretation: The design is not a causal shock design. It is a macro-finance asset-pricing exercise that links portfolio returns to cross-country exposure heterogeneity and changes in global expectations.

4.2 What the model must explain

- Traditional carry is positive in the full sample but lower after 2008.
- Slope carry is near zero in the full sample, negative before 2008, and strongly positive after 2008.
- Expected global growth and inflation decline after 2008.
- Countries have heterogeneous exposures to global expected growth and inflation.
- Portfolio-level exposures change when countries are resorted into portfolios after the regime shift.

Interpretation: The paper's identification comes from fitting all these facts simultaneously. A model with only global growth exposures cannot explain the slope carry regime shift; global inflation exposures are needed.

4.3 Limitations

- The model abstracts from many frictions in international bond and currency markets.
- The baseline analysis uses complete markets and representative agents.
- The pre/post-2008 break is economically motivated but not a randomized event.
- The number of G10 countries is small, so portfolio-level evidence is informative but limited.
- Some robustness and calibration details are placed in the Internet Appendix.

Interpretation: The paper's main contribution is not a micro-level causal estimate. It is a coherent mechanism linking global inflation risk to the previously hidden profitability of slope carry.

5 Tables and figures

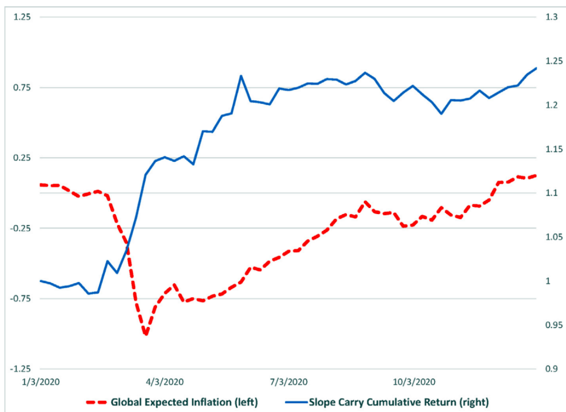
Visuals are directly extracted from the paper and grouped compactly.

5.1 Figure 1: pandemic-era global inflation expectations and slope carry

Read:

- The figure plots GDP-weighted 5-year breakeven inflation and the cumulative slope carry return around 2020.
- Inflation expectations fall sharply in early 2020, and the slope carry becomes very profitable.
- As inflation expectations recover, the slope carry stops earning such large returns.

Economic message: This is motivating evidence for the model: slope carry profitability rises when expected global inflation is low.



5.2 Tables 1 and 2: traditional carry and slope carry returns

Read:

- Table 1 shows that traditional carry earns a positive high-minus-low excess return over the whole sample.
- The traditional carry premium is strong pre-2008 but much weaker post-2008.
- Table 2 shows that slope carry is close to zero over the whole sample.
- Slope carry is slightly negative pre-2008 and strongly positive post-2008, with a high-minus-low return of about 4.75%.

Economic message: The slope carry return is concealed in long samples because the pre- and post-2008 premia have opposite signs.

Table 1

Traditional Carry. The table reports the excess returns associated to borrowing at th 3 months interest rate of the US and investing in 3 months bonds of a GDP-weighte portfolio of countries with low (1), medium (2), and high (3) interest rates. The colum label "3-1" reports the average return from being long portfolio 3 and short portfoli 1. Portfolios are rebalanced every month. Returns are in gross units. The analysi is conducted over three samples: 1/1995-12/2020 ("Whole sample"), 1/1995-7/200 ("Pre-08/2008"), and 8/2008-12/2020 ("Post-08/2008"). Numbers in square bracket denote standard errors. Numbers in parentheses refer to the frequency with which country belongs to a specific portfolio.

	1 (Low)	2	3 (High)	3-1 (High–Low)
Whole Sample				
Mean	−1.99	0.34	3.22	5.21***
				[1.92]
Sharpe Ratio	−0.24	0.04	0.32	0.53
Pre-08/2008				
Mean	−3.37	2.01	5.59	8.96***
				[2.47]
Sharpe Ratio	−0.37	0.29	0.76	0.99
Recurrent countries:	Jpn (100%)	Can (77%)	NZ (91%)	
	Swi (100%)	Swe (62%)	Aus (90%)	
	Ger (40%)	UK (34%)	UK (66%)	
Post-08/2008				
Mean	−0.48	−1.47	0.64	1.12
				[1.39]
Sharpe Ratio	−0.06	−0.18	0.05	0.11
Recurrent countries:	Swi (95%)	UK (95%)	NZ (100%)	

Table 1: Traditional carry

Table 2

Slope Carry. The table reports the excess returns associated to borrowing at the 3 months interest rate of the US and investing in the 10 year bonds of a GDP-weighted portfolio of countries with flatter (1), medium (2), and steeper (3) yield curves. The column label "3-1" reports the average return from being long portfolio 3 and short portfolio 1. Portfolios are rebalanced every month. Returns are in gross units. The analysis is conducted over three samples: 1/1995-12/2020 ("Whole sample"), 1/1995-7/2008 ("Pre-08/2008"), and 8/2008-12/2020 ("Post-08/2008"). Numbers in square brackets denote standard errors. Numbers in parentheses refer to the frequency with which a country belongs to a specific portfolio.

	1 (Flatter)	2	3 (Steeper)	3-1 (Steep-flat)
Whole Sample				
Mean	4.69	2.22	6.58	1.89
				[2.20]
Sharpe Ratio	0.46	0.22	0.69	0.20
Pre-08/2008				
Mean	6.55	3.95	5.80	−0.75
				[2.20]
Sharpe Ratio	0.66	0.38	0.53	−0.07
Recurrent countries:	UK (83%)	Ger (55%)	Swe (59%)	
	NZ (76%)	Swi (43%)	Jpn (56%)	
	Aus (71%)	Jpn (42%)	Swi (49%)	
Post-08/2008				
Mean	2.67	0.34	7.42	4.75***
				[2.08]
Sharpe Ratio	0.26	0.03	0.88	0.51
Recurrent countries:	Jpn (75%)	Swi (53%)	UK (62%)	

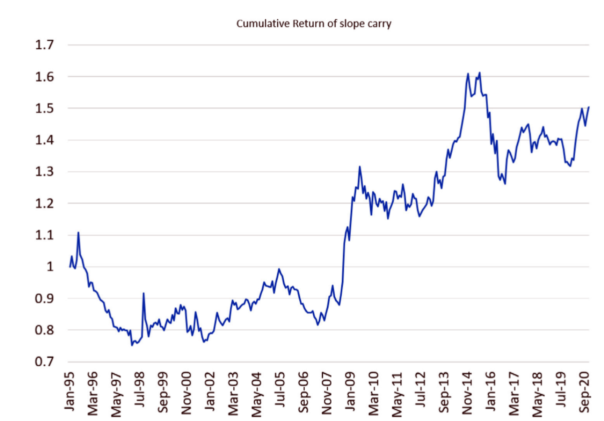
Table 2: Slope carry

5.3 Figure 2 and Table 3: cumulative slope carry and exchange-rate component

Read:

- Figure 2 shows cumulative slope carry returns from 1995 to 2020.
- The strategy loses value before 2008 and rises sharply after 2008.
- Table 3 shows that exchange-rate depreciation contributes positively to the post-2008 slope carry return.

Economic message: The slope-carry regime shift is visible not only in average returns but also in the cumulative time-series path and exchange-rate component.



eturn of the Slope Carry. The figure depicts the cumulative return of a slope carry investment strategy; that is, a strategy lon

Figure 2: cumulative slope carry

Table 3

The Role of Exchange Rates. The table reports the excess returns associated to borrowing at the 3 months interest rate of the US and investing in the 10 year bonds of a GDP-weighted portfolio of countries with flatter (1), medium (2), and steeper (3) yield curves. The column label "3-1" reports the average return from being long portfolio 3 and short portfolio 1. Portfolios are rebalanced every month. Returns are in gross units. The analysis is conducted over three samples: 1/1995-12/2020 ("Whole sample"), 1/1995-7/2008 ("Pre-08/2008"), and 8/2008-12/2020 ("Post-08/2008"). Numbers in square brackets denote standard errors. Numbers in parentheses refer to the frequency with which a country belongs to a specific portfolio.

	Traditional carry		Slope carry	
	Whole	Post-08	Whole	Post-08
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1	1.39	−1.53	0.92	3.40
	(1.83)	(1.28)	(2.34)	(2.26)

Table 3: exchange-rate component

5.4 Figure 3 and Tables 4–5: global expectations and exposures

Read:

- Figure 3 shows a post-2008 decline in expected global GDP growth and expected global inflation.
- Table 4 shows heterogeneous country exposures to global expected growth and inflation.
- Japan has a high exposure to global expected growth; the UK, Sweden, and Switzerland have relatively high exposure to global expected inflation.
- Table 5 shows statistically meaningful differences in exposures across Australia, Japan, and the UK.

Economic message: The model needs two forms of cross-country heterogeneity: exposure to growth news for traditional carry and exposure to inflation news for slope carry.

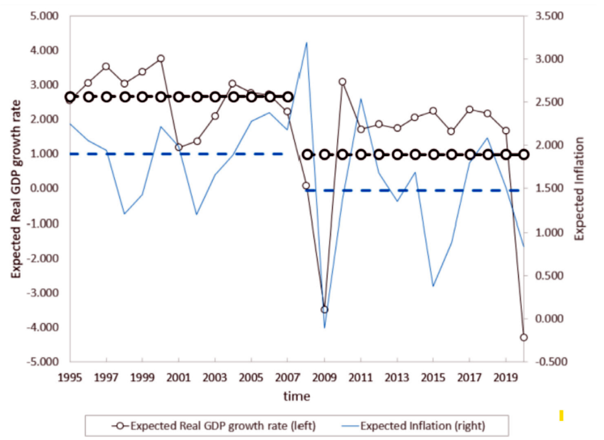


Figure 3: expected global inflation and GDP growth

Table 4

Expectations Exposures. Exposures of each country's expected GDP growth rate and expected inflation to GDP weighted expectations of GDP and inflation. The first panel reports the estimates of $\beta_{i,y}$ in Eq. (1). The second panel reports the estimates of $\beta_{i,\pi}$ in Eq. (2). The numbers in parentheses underneath the estimated coefficients are standard errors.

Exposures to expected GDP growth

	NZL	NOR	AUS	SWI	SWE	UK	US	CAN	GER	JPN
$\beta_{i,y}$	0.353	0.492	0.532	0.541	0.606	0.908	0.923	0.976	0.997	1.422
	(0.174)	(0.098)	(0.106)	(0.067)	(0.152)	(0.159)	(0.055)	(0.079)	(0.117)	(0.191)

Exposures to Expected Inflation

	NOR	NZL	AUS	CAN	JPN	GER	US	UK	SWI	SWE
$\beta_{i,\pi}$	0.233	0.568	0.591	0.627	0.670	0.821	1.107	1.240	1.454	1.762
	(0.125)	(0.218)	(0.148)	(0.087)	(0.186)	(0.066)	(0.079)	(0.266)	(0.242)	(0.226)

Table 4: expectation exposures

Table 5

Differences of Expectations Exposures. This table reports differences of exposures o expected GDP growth rate (left) and expected inflation (right) between Australia, Japan and UK. Each entry represents the difference between the exposures of the country in each column and the country in the row. Numbers in parentheses are standard errors One, two, and three stars represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Expected GDP growth				*Expected inflation*		
	AUS	JPN	UK		AUS	JPN	UK
AUS	–	0.889***	0.376**	**AUS**	–	0.079	0.649**
		(0.153)	(0.159)			(0.211)	(0.276)
JPN		–	−0.514*	**JPN**		–	0.570**
			(0.304)				(0.230)
UK			–	**UK**			–

5.5 Tables 6 and 7: calibration and simulated cross-country moments

Read:

- Table 6 reports the baseline quarterly calibration of preferences, growth, inflation, and long-run shock processes.
- Table 7 compares empirical cross-country moments with simulated moments.
- The model matches key averages and cross-sectional dispersion reasonably well, though with some limitations in volatility and correlation moments.

Economic message: The calibration is designed to discipline the model before using it to interpret carry returns.

Table 6

Calibration. This table reports the value of our parameters under our baseline calibration. Some parameters are calibrated to be within the confidence intervals of their counterpart estimated in the data. HAC-corrected standard errors are reported in the parentheses. Other parameters are calibrated to match cross sectional averages (Avg.$_i$) in the data. Empirical estimates are from the specification detailed in Eq. (3). Our data set is detailed in Section 2.

Description	Parameter	Value	Estimate/ moment
Subjective discount factor	δ	0.997	Avg.$_i$ $\left[E(r^f)\right]$
Risk Aversion	γ	10	$E(carry^S)$
Cross-country average consumption growth	$\bar{\mu}_c$	0.49%	Avg.$_i$ $[\Delta c]$
Volatility of cons growth short-run shock	σ_c	0.46%	Avg.$_i$ $[\sigma(\Delta c)]$
Volatility of cons growth long-run shock	σ_{xc}	0.11%	Avg.$_i$ $\left[ACF_1(\Delta c)\right]$
Autocorr. cons growth long-run risk	ρ_c	0.810	0.601 (0.145)
Cross-country average inflation growth	$\bar{\mu}_\pi$	0.25%	Avg.$_i$ $[\pi]$
Volatility of inflation short-run shock	σ_π	0.55%	Avg.$_i$ $[\sigma(\pi)]$
Volatility of inflation long-run shock	$\sigma_{x\pi}$	0.11%	Avg.$_i$ $\left[ACF_1(\pi)\right]$
Autocorr. inflation long-run risk	ρ_π	0.988	0.916 (0.040)
Cons growth/inflation long-run feedback	$\rho_{c\pi}$	−0.050	−0.019 (0.039)

Table 6: calibration

Table 7

Heterogeneous Exposure and Cross-sectional Moments. The table reports cross sectional averages (Avg.$_i$) and cross sectional coefficients of variation (StDev$_i$/Avg.$_i$) for several moments of interest. The column 'Value' reports our point estimates computed using the data set described in Section 2. We report the associated HAC-adjusted standard errors under the column 'Std Err'. The entries for the column 'Model' are obtained by simulating 1000 short samples comprising 100 quarterly observations. Simulated data are time aggregated at the annual frequency. All parameters are set to their benchmark values reported in Table 6. For $ACF_1(\Delta c)$, $ACF_1(\pi)$ and $Corr(\Delta c, \pi)$, we report StDev$_i$ rather than StDev$_i$/Avg.$_i$.

	Avg.$_i$			StDev$_i$/Avg.$_i$		
	Value	(Std err)	Model	Value	(Std err)	Model
β_c	0.91	(0.08)	0.65	0.39	(0.11)	0.36
$E(\Delta c)$	2.21	(0.18)	2.60	0.35	(0.04)	0.19
$\sigma(\Delta c)$	1.08	(0.18)	1.02	0.27	(0.06)	0.16
$ACF_1(\Delta c)$	0.35	(0.12)	0.08	0.19	(0.04)	0.21
β_π	0.96	(0.11)	0.96	0.47	(0.17)	0.40
$E(\pi)$	1.66	(0.13)	1.20	0.43	(0.05)	0.59
$\sigma(\pi)$	0.94	(0.12)	1.75	0.21	(0.04)	0.25
$ACF_1(\pi)$	0.20	(0.12)	0.40	0.18	(0.04)	0.22
$Corr(\Delta c, \pi)$	−0.23	(0.10)	−0.12	0.34	(0.05)	0.19

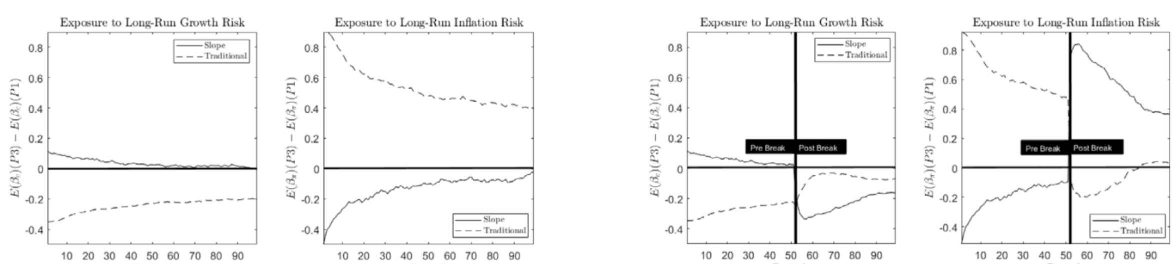
Table 7: cross-sectional moments

5.6 Figures 4–5 and Table 8: simulated portfolio exposures and returns

Read:

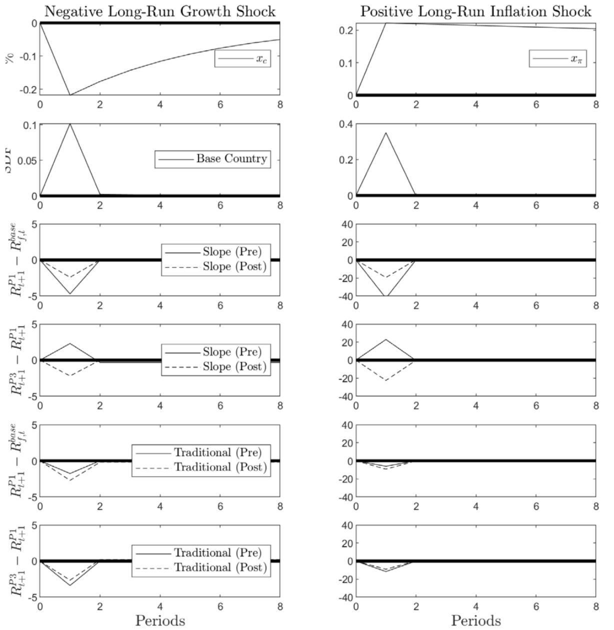
- Figure 4 shows that portfolio-level exposures differ for traditional and slope carry and change after the break.
- Figure 5 shows impulse responses to global growth and inflation news shocks.
- Table 8 reports simulated moments for traditional and slope carry.
- The model produces a nearly zero full-sample slope carry premium and a strongly positive post-2008 slope carry premium.

Economic message: The slope carry premium changes sign because the slope-sorted portfolios load differently on global inflation risk across regimes.



(a) Without Break (b) With Break

nulated Portfolio-Level Exposures. This figure shows differences in simulated portfolio-level exposures to long-run growth risk (β_c, left panels) and expect ws shocks (β_π, right panels). In panel (a), there is no break in expected growth and inflation. In panel (b), a break suddenly reduce both expected growth and litional carry, portfolio P1 (P3) comprises low-interest rate (high-interest rate) countries. For the slope carry, portfolio P1 (P3) comprises flatter-yield curve (ste



e to Global News Shocks. This figure shows portfolio-level impulse response functions for the Pre-Break (solid line) and P

Figure 5: portfolio impulse responses

Table 8

Simulated Moments. This table reports both empirical and simulated moments for both the traditional and the slope carry strategies. All moments are (i) computed as GDP weighted averages within each portfolio, (ii) annualized, and (iii) multiplied by 100 (except for β_c and β_π). For the traditional (slope) carry, the "high" portfolio, P3 comprises countries with high short-term interest rates (steeper yield curve slopes). The opposite is true for the "low" portfolio, P1. The entries for the moments are based on 1000 simulations of 100 quarters. All parameters are set to their benchmark values reported in Table 6. $E(\Delta FX)$ refers to the average exchange rate depreciation and $E(\pi)$ refers to the average inflation rate. $E(\beta_c)$ ($E(\beta_\pi)$)) measures the portfolio-level exposure to global expected consumption growth (inflation). The numbers in parentheses denote standard errors.

	Traditional carry		Slope carry	
	Data	Model	Data	Model
$E(carry)$ (Full Sample)	5.21	2.75	1.89	2.36
	(1.92)		(2.20)	
$E(carry)$ (Post-08/08)	1.12	1.48	4.75	7.65
	(1.39)		(2.08)	
$E(sortingvar)$ P3 - $E(sortingvar)$ P1	3.83	1.91	1.44	1.75
	(0.96)		(0.30)	
$E(sortingvar)$ P3 - $E(sortingvar)$ P1 (Post-08/08)	2.65	1.52	1.08	1.98
	(1.64)		(0.25)	
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1	1.39	0.85	0.92	2.32
	(1.83)		(2.34)	
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1 (Post-08/08)	−1.53	−0.02	3.40	4.03
	(1.28)		(2.26)	
$E(\beta_c)$ P3 - $E(\beta_c)$ P1	−0.63	−0.17	0.11	−0.06
$E(\beta_c)$ P3 - $E(\beta_c)$ P1 (Post-08/08)	−0.56	−0.06	−0.27	−0.20
$E(\beta_\pi)$ P3 - $E(\beta_\pi)$ P1	−0.11	0.29	0.00	0.16
$E(\beta_\pi)$ P3 - $E(\beta_\pi)$ P1 (Post-08/08)	−0.35	−0.07	0.19	0.55
$E(\pi)$ P3 - $E(\pi)$ P1	2.13	1.42	−0.33	−1.41
	(0.65)		(0.96)	
$E(\pi)$ P3 - $E(\pi)$ P1 (Post-08/08)	1.06	1.23	1.23	−1.33

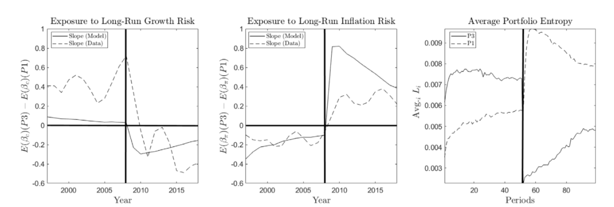
Table 8: simulated moments

5.7 Figure 6 and Table 9: model-data exposures and entropy

Read:

- Figure 6 compares simulated and empirical portfolio-level exposures to growth and inflation risk.
- It also plots average portfolio entropy for the extreme slope portfolios.
- Table 9 shows that differences in entropy across extreme portfolios forecast slope carry returns.

Economic message: Portfolio composition makes risk exposure and entropy time-varying even though country-level exposures are fixed in the baseline model.



pe Carry: Portfolio-Level Exposures for Model and Data. The first two panels of this figure show differences in portfolio-level exposures to long-run growtl and expected global inflation news shocks (β_π, right panel) for the slope carry. The solid lines are obtained using simulated data. Our quarterly calibration i The dashed lines show the empirical estimates obtained using the data set of Section 2. Portfolio P1 (P3) comprises flatter-yield curve (steeper-yield curve)

Figure 6: model and data exposures

Table 9

Simulated Entropy and Predictability. This table reports average simulated moments based on 1000 simulations of 100 quarters. For the slope carry, the "high" portfolio P3, comprises countries with high steeper yield curve slopes. The opposite is true for the "low" portfolio, P1. We run the following regression: $\log RFX^{P3}_{t+1} - \log RFX^{P1}_{t+1} = c_0+\beta F_t+resid_{t+1}$, where F_t is a forecasting variable. As in Lustig et al. (2019b), F_t refers to the difference across portfolios of either (i) the level of the yield curves ($r^1_{P3,t} - r^1_{P1,t}$ column 'YC Level') or (ii) the slope of the yield curves ($slope^\infty_{P3,t} - slope^\infty_{P1,t}$). We report both the average point estimate of β and its average standard error across repetitions. 'Entropy' refers to the annualized percentage of the average difference of entropies of the permanent components of the pricing kernels in the two extreme portfolios ($L^{P1} - L^{P3}$). These figures are annualized and multiplied by 100. All parameters are set to their benchmark values reported in Table 6.

	YC level	YC slope	Entropy
Full Sample			
Point Est.	−5.14	6.15	0.80
St. Err.	(4.66)	(6.82)	–
Post-08/08			
Point Est.	−12.11	14.84	2.38

Table 9: entropy and predictability

5.8 Table 10 and Figure 7: the 1975–1985 inflation episode

Read:

- Table 10 studies slope carry during 1975–1985, a period with large swings in inflation.

- The slope carry earns a high return in the 1975–1979 subsample and a negative return in the 1979–1980 subsample.
- Figure 7 shows that profitability rises when expected inflation is relatively low and declines when expected inflation is high.

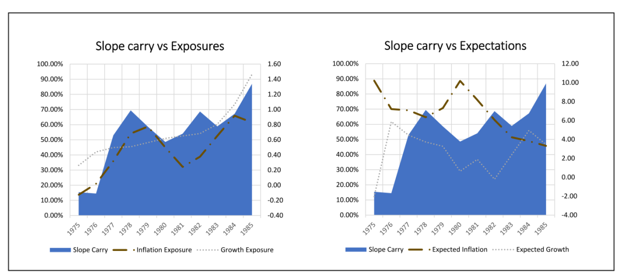
Economic message: The inflation-risk mechanism applies beyond the post-2008 episode; it also helps interpret the high-inflation era of the 1970s and early 1980s.

Table 10

Slope Carry: 1975–1985. This table reports the excess returns associated to borrowin at the 3 months interest rate of the US and investing in the 10 year bonds of a GDP weighted portfolio of countries with flatter (1), medium (2), and steeper (3) yiel curves. The column label "3-1" reports the average return from being long portfolio 3 and short portfolio 1. Portfolios are rebalanced every month. Returns are in log-units The rows labeled "$E[\beta_\pi](P3) - E[\beta_\pi](P1)$" and "$E[\beta_c](P3) - E[\beta_c](P1)$" denote th spread of the inflation and growth exposures between portfolios 3 and 1. The inflatio and growth exposures β_π and β_c are estimated on the sample 1975–1990. The number in square brackets are t-statistics.

	1 (Flatter)	2	3 (Steeper)	3-1 (Steep-flat)
Sample: 12/1974–12/1985				
Mean	−12.06%	−9.23%	−4.32%	7.74%
				[2.00]
Sharpe Ratio	−0.79	−0.63	−0.25	0.49
Recurrent countries:	NZ (95%)	Ger (57%)	Jpn (71%)	
	Swe (56%)	Aus (50%)	Nor (61%)	
	Can (52%)	Swi (41%)	UK (41%)	
$E[\beta_\pi](P3) - E[\beta_\pi](P1)$	1.13	1.03	1.21	0.08
$E[\beta_c](P3) - E[\beta_c](P1)$	0.42	0.59	0.56	0.14
Sample: 12/1974–12/1979				
Mean	−8.2%	−5.89%	3.08%	11.28%
				[2.28]
Sharpe Ratio	−0.86	−0.49	0.2	0.69
Recurrent countries:	NZ (100%)	Nor (67%)	UK (66%)	
	Can (79%)	Aus (66%)	Swi (56%)	
	Swe (43%)	Ger (56%)	Jpn (54%)	
$E[\beta_\pi](P3) - E[\beta_\pi](P1)$	1.14	0.96	1.28	0.14
$E[\beta_c](P3) - E[\beta_c](P1)$	0.48	0.59	0.54	0.06
Sample: 12/1979–12/1980				
Mean	−5.91%	−34.96%	−11.78%	−5.88%
				[-0.73]
Sharpe Ratio	−0.41	−1.9	−0.46	−0.3
Recurrent countries:	UK (92%)	Aus (69%)	Jpn (100%)	
	NZ (69%)	Swi (69%)	Nor (100%)	
	Can (46%)	Ger (62%)	Can (38%)	
$E[\beta_\pi](P3) - E[\beta_\pi](P1)$	1.74	0.73	1.15	−0.59
$E[\beta_c](P3) - E[\beta_c](P1)$	0.46	0.54	0.57	0.1
Sample: 12/1980–12/1985				
Mean	−18.07%	−7.55%	−9.66%	8.41%
				[2.27]

Table 10: 1975–1985 slope carry



Slope Carry, Exposures, and Expectations: 1975–1985. The left panel reports the cumulative Slope Carry Excess return (shaded area), the cumulative diff exposures between the extreme portfolios (dash-dot line), and the cumulative difference of GDP growth exposures between the extreme portfolios (dotted line).

Figure 7: exposures and expectations, 1975–1985

5.9 Tables 11 and 12: IES and demand-shock extensions

Read:

- Table 11 shows that the main results survive when the intertemporal elasticity of substitution is set above one.
- Table 12 adds global demand shocks and shows that the main carry moments are preserved.
- Demand shocks help reduce the variance contribution of inflation shocks to local yield volatility, bringing the model closer to external evidence.

Economic message: The slope-carry mechanism is robust to richer preference and shock structures.

Table 11
The Role of IES. This table reports both empirical and simulated moments for botl the traditional and the slope carry strategies. All moments are (i) computed as GDP weighted averages within each portfolio, (ii) annualized, and (iii) multiplied by 10((except for β_c and β_π). For the traditional (slope) carry, the "high" portfolio, P3 comprises countries with high short-term interest rates (steeper yield curve slopes) The opposite is true for the "low" portfolio, P1. The entries for the moments ar based on 1000 simulations of 100 quarters. All parameters are set to their benchmarl values reported in Table 6, and the IES is also allowed to be 2. $E(\Delta FX)$ refers to th average exchange rate depreciation. $E(\beta_c)$ ($E(\beta_\pi)$) measures the portfolio-level exposur to global expected consumption growth (inflation). The numbers in parentheses denot standard errors.

	Traditional carry			Slope carry		
	Data	IES=1	IES=2	Data	IES=1	IES=2
$E(carry)$ (Full Sample)	5.21	2.75	2.07	1.89	2.36	3.85
	(1.92)			(2.20)		
$E(carry)$ (Post-08/07)	1.12	1.48	1.11	4.75	7.65	9.42
	(1.39)			(2.08)		
$E(sortingvar)$ P3 - $E(sortingvar)$ P1	3.83	1.91	1.80	1.44	1.75	1.79
	(0.96)			(0.30)		
$E(sortingvar)$ P3 - $E(sortingvar)$ P1 (Post-08/08)	2.65	1.52	1.44	1.08	1.98	2.16
	(1.64)			(0.25)		
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1	1.39	0.85	0.28	0.92	2.32	1.74
	(1.83)			(2.34)		
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1 (Post-08/08)	−1.53	−0.02	−0.31	3.40	4.03	3.03
	(1.28)			(2.26)		
$E(\beta_c)$ P3 - $E(\beta_c)$ P1	−0.63	−0.17	−0.16	0.11	−0.06	−0.06
$E(\beta_c)$ P3 - $E(\beta_c)$ P1 (Post-08/08)	−0.56	−0.06	−0.04	−0.27	−0.20	−0.21
$E(\beta_\pi)$ P3 - $E(\beta_\pi)$ P1	−0.11	0.29	0.28	0.00	0.16	0.29

Table 11: role of the IES

Table 12
The Role of Demand Shocks. This table reports both empirical and simulated moment for both the traditional and the slope carry strategies. All moments are (i) compute as GDP-weighted averages within each portfolio, (ii) annualized, and (iii) multiplie by 100 (except for β_c and β_π). For the traditional (slope) carry, the "high" portfolio P3, comprises countries with high short-term interest rates (steeper yield curve slopes) The opposite is true for the "low" portfolio, P1. The entries for the moments are base on 1000 simulations of 100 quarters. All parameters are set to their benchmark value reported in Table 6, except the IES that is set to 2. When the demand shock is present we set $\sigma_d = 5e^{-4}$ and $\rho_d = .9742$. When we include a downward jump in the deman process, we set it equal to -1StDev(x_d). $E(\Delta FX)$ refers to the average exchange rat depreciation. $Slope$ and hpr^∞ refer to the slope and the holding period return of a infinite-maturity bond, respectively. The share of volatility refers to the simple averag of the country-level shares. The numbers in parentheses denote standard errors.

Panel A: International Moments

	Traditional Carry			Slope Carry		
Demand shock	yes	yes	no	yes	yes	no
Demand shock downward jump	yes	no	–	yes	no	–
$E(carry)$ (Full Sample)	1.42	1.42	2.07	0.98	0.98	3.85
$E(carry)$ (Post-08/07)	0.47	0.47	1.11	5.50	5.50	9.42
$E(sortingvar)$ P3 - $E(sortingvar)$ P1	1.74	1.74	1.80	1.66	1.66	1.79
$E(sortingvar)$ P3 - $E(sortingvar)$ P1 (Post-08/08)	1.43	1.43	1.44	1.83	1.83	2.16
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1	−0.30	−0.30	0.28	1.53	1.53	1.74
$E(\Delta FX)$ P3 - $E(\Delta FX)$ P1 (Post-08/08)	−0.93	−0.93	−0.31	2.58	2.58	3.03

Panel B: Local Moments

Share of volatility due to inflation	hpr^∞	$Slope$	ΔFX
With demand shock	9.6%	58.6%	25.2%
Without demand shock	79.2%	81.0%	28.5%

Table 12: role of demand shocks

6 Short summary

- The paper studies traditional currency carry and slope carry in G10 countries.
- Traditional carry is sorted on short-rate levels; slope carry is sorted on yield-curve slopes.
- Traditional carry remains positive but weakens after 2008.
- Slope carry is close to zero in the full sample, negative before 2008, and strongly positive after 2008.
- Expected global growth and expected global inflation decline after 2008.
- Countries have heterogeneous exposures to global expected growth and global expected inflation.
- Growth exposure explains traditional carry; inflation exposure explains slope carry.
- The model reproduces the sign change in slope carry through changes in portfolio-level inflation exposure.
- The mechanism also helps explain the slope carry during the 1975–1985 inflation episode.
- The broad lesson is that yield-curve slope carry contains an inflation-risk premium that is concealed in long samples.

7 Conclusion

7.1 Core conclusion

Andrews, Colacito, Croce, and Gavazzoni show that the slope carry premium is concealed over long samples because it is regime-dependent. It is negative or weak when expected global inflation is high, but strongly positive when expected global inflation is low. The post-2008 environment reveals this hidden inflation-risk premium.

7.2 Economic intuition

Countries differ in how much their expected inflation responds to global inflation news. When global expected inflation is low, countries with high inflation exposure have steeper yield curves and require larger inflation risk premia on long-term bonds. The slope carry strategy goes long these steeper-yield-curve countries and earns a positive premium.

7.3 Takeaway

The paper adds a global inflation-risk channel to the international carry literature. Traditional carry is mostly about growth risk and short-rate levels; slope carry is mostly about inflation risk and yield-curve slopes. Understanding both requires looking at the macro regime and the cross-section of country exposures.